Moongyu Jeon

Ph.D. Student in Artificial Intelligence, Yonsei University

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RESEARCH INTERESTS

My research focuses on theoretically understanding diffusion language models and, more broadly, on developing future generative language models whose behavior and capabilities can be understood through principled theory.

EDUCATION

Yonsei University, Seoul, Korea — Mar. 2025 - Present
Ph.D. Student in Artificial Intelligence
Advisor: Prof. Albert No

Yonsei University, Seoul, Korea — Mar. 2020 - Feb. 2025
B.S. in Mathematics · GPA: 4.20/4.30
Department Valedictorian (Rank 1/42)

PUBLICATIONS

** denotes equal contribution.*

Low-Confidence Remasking Traps Flexibility: Realizing Arbitrary-Order Potential for Diverse Rollouts in Diffusion LLMs
Moongyu Jeon*, Dongjae Jeon*, Bumjun Kim, Mingyu Kim, Albert No
Conference on Neural Information Processing Systems **(NeurIPS)** 2026 Workshops: BeNTo and DiffuLM
arXiv

A Theoretical Analysis of Why Masked Diffusion Models Mitigate the Reversal Curse
Moongyu Jeon*, Sangwoo Shin*, Bumjun Kim, Kyelim Lee, Albert No
Conference on Neural Information Processing Systems **(NeurIPS)** 2026
arXiv

Preserve-Then-Quantize: Balancing Rank Budgets for Quantization Error Reconstruction in LLMs
Yoonjun Cho*, Dongjae Jeon*, Soeun Kim, **Moongyu Jeon**, Albert No
International Conference on Machine Learning **(ICML)** 2026
arXiv | Project | Code

DAPD: Dependency-Aware Parallel Decoding via Attention for Diffusion LLMs
Bumjun Kim*, Dongjae Jeon*, **Moongyu Jeon***, Albert No
International Conference on Machine Learning **(ICML)** 2026
arXiv | Project | Code

Information-Theoretic Discrete Diffusion
Moongyu Jeon, Sangwoo Shin, Dongjae Jeon, Albert No
Conference on Neural Information Processing Systems **(NeurIPS)** 2025
arXiv | Code

RESEARCH EXPERIENCE

Student Researcher, AI-ISL, Yonsei University — Mar. 2025 - Present

TEACHING EXPERIENCE

Teaching Assistant, CAS3230 Introduction to Mathematics for Deep Learning — Fall 2026
Teaching Assistant, CAS3230 Introduction to Mathematics for Deep Learning — Fall 2025
Teaching Assistant, CAS4108 Data Privacy — Spring 2025

ACADEMIC SERVICE

Conference on Neural Information Processing Systems **(NeurIPS)** — **Top Reviewer**, 2026
International Conference on Learning Representations **(ICLR)** — Reviewer, 2027
Conference on Neural Information Processing Systems **(NeurIPS)** Workshop: BeNTo — Reviewer, 2026